Universal Inquiry

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v1

Chapter 1

Logic

A variable is an object. If a, b are variables, then $a \in b$ is a formula. It is true if a is in b, and false otherwise. If p is a formula, then $\neg p$ is a formula. It is true if p is false, and false otherwise. If p, q are formulas, then $p \wedge q$ is a formula. It is true if p, q are true, and false otherwise. If x is a variable, and p is a formula, then $\exists x\,(p)$ is a formula. It is true if there exists x such that p is true, and false otherwise.

Notation 1.1.

$$p \vee q \equiv \neg(\neg p \wedge \neg q)$$

Remark 1.1. This says "it is not the case that both p and q fail".

Notation 1.2.

$$p \Rightarrow q \equiv \neg(p \wedge \neg q)$$

Remark 1.2. This says "it is not the case that p works but q fails".

Notation 1.3.

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

Notation 1.4.

$$p \oplus q \equiv \neg(p \Leftrightarrow q)$$

Remark 1.3. This says "it is not the case that p and q work the same".

Notation 1.5.

$$\forall x\,(p) \equiv \neg\exists x\,(\neg p)$$

Remark 1.4. This says "no counterexample exists".

Notation 1.6.

$$X = Y \Leftrightarrow \forall a\,(a \in X \Leftrightarrow a \in Y) \wedge \forall B\,(X \in B \Leftrightarrow Y \in B)$$

Remark 1.5. This says "X and Y look the same from the inside and outside".

Axiom 1.1.

$$\forall A, B\,(\forall x\,(x \in A \Leftrightarrow x \in B) \Rightarrow A = B)$$

Axiom 1.2.

$$\forall B, \exists U, \forall c\,(c \in U \Leftrightarrow \exists P \in B\,(c \in P))$$

Definition 1.1.

$$U = \bigcup B \Leftrightarrow \forall c\,(c \in U \Leftrightarrow \exists P \in B\,(c \in P))$$

Definition 1.2.

$$A \cup B = \bigcup \{A, B\}$$

Definition 1.3.

$$I = \bigcap B \Leftrightarrow \forall c\,(c \in I \Leftrightarrow \forall P \in B\,(c \in P))$$

Definition 1.4.

$$A \cap B = \bigcap \{A, B\}$$

Axiom 1.3.

$$\forall X, \exists P, \forall S\,(S \in P \Leftrightarrow \forall s \in S\,(s \in X))$$

Axiom 1.4.

$$\exists I\,(\varnothing \in I \land \forall i \in I\,(i \cup \{i\} \in I))$$

Axiom 1.5.

$$\forall X \neq \varnothing, \exists x \in X\,(x \cap X = \varnothing)$$

Notation 1.7.

$$\begin{aligned} \operatorname{free}(a \in b) &= \{a, b\} \\ \operatorname{free}(\neg p) &= \operatorname{free}(p) \\ \operatorname{free}(p \land q) &= \operatorname{free}(p) \cup \operatorname{free}(q) \\ \operatorname{free}(\exists x\,(p)) &= \operatorname{free}(p) \setminus \{x\} \end{aligned}$$

Axiom 1.6. Let $B \notin \operatorname{free}(p)$.

$$\forall A\,(\forall x \in A, \exists! y\,(p) \Rightarrow \exists B, \forall y\,(y \in B \Leftrightarrow \exists x \in A\,(p)))$$

Definition 1.5.

$$A \times B = \{(a, b) \mid a \in A \land b \in B\}$$

Definition 1.6.

$$Y^X = \{f \subseteq X \times Y \mid \forall x \in X, \exists! y \in Y\,((x, y) \in f)\}$$